

Travelling Beam Technical Datasheet

Generic technical data and inspection reference for travelling beams used in theatre, venue and entertainment lifting or suspension arrangements.

Purpose of this document

This datasheet is a generic template for portal use. It should be completed or amended using manufacturer information, installation records, structural calculations, inspection reports and site-specific limitations. It is not a substitute for the manufacturer manual, competent-person assessment or written scheme of examination.

1. Equipment overview

A travelling beam is a movable beam or beam-mounted lifting/suspension arrangement used to provide a travelling load path or movable suspension point. In theatre and venue environments it may be used with trolleys, hoists, bars, scenic loads, curtains, speakers or other production equipment. The beam, supports, end stops, trolleys, suspension points, fixings and any lifting appliance fitted to it must be assessed as one complete load path.

2. Technical data record

Field	Information / site-specific entry
Equipment name	Travelling Beam
Manufacturer / supplier	To be completed from manufacturer or installation records
Model / type / reference	To be completed
Location / asset number	To be completed
Beam type / section	To be completed - e.g. steel runway beam, aluminium beam or proprietary track/beam system
SWL / WLL	To be completed and verified for the beam, trolley, support structure and lifting appliance
Maximum span / travel length	To be completed from design or installation information
Approved load direction	To be completed - normally vertical loading only unless otherwise verified
Compatible trolley / hoist	To be completed - trolley type, wheel profile, hoist rating and attachment method
Support structure / fixings	To be completed - parent structure, brackets, hangers, anchors and support points
End stops / travel limits	To be completed - end stops, buffers, lock-offs or parking arrangements
Inspection interval	To be completed in line with written scheme, site risk assessment and competent-person advice
Documents to hold	Manufacturer manual, installation drawings, calculations, certificates, test records and examination reports

3. Regulatory and good-practice context

- LOLER is relevant where the travelling beam forms part of lifting equipment or supports a load during lifting operations. The complete load path should be suitable, strong, stable, marked with any required WLL/SWL information and subject to thorough examination where required.
- PUWER applies because the travelling beam and associated equipment are work equipment. They must be suitable for the intended use, maintained, inspected where deterioration could create risk and used by trained/competent persons.
- BS EN 17206 should be considered where the travelling beam forms part of stage or production machinery, powered movement, lifting systems or entertainment load-bearing systems.
- ABTT Technical Standards and recognised industry good practice should be used to support venue-specific inspection, maintenance, record keeping and competent operation.

4. Suitability and installation considerations

- Confirm the travelling beam is suitable for the intended load, load movement, duty, span, support spacing and environment.
- The rated capacity of the complete arrangement is limited by the lowest-rated component in the load path, including beam, trolley, fixings, support steelwork, hoist, shackles and attachments.
- Check whether the beam is intended for static suspension, manual travel, powered travel or lifting-machine use. Do not assume suitability for a new use without assessment.
- End stops, anti-derailment features, buffers, parking locks and travel limits should be fitted where required and kept in serviceable condition.
- Loads should be applied in the approved direction. Side loading, shock loading, impact, twisting or dragging loads can significantly increase risk.
- The parent structure must be verified as suitable for the imposed loads, including dynamic effects and travelling load positions.

5. Inspection checklist

Area	Inspection / verification points	Status
Identification and rating	Check asset identification, SWL/WLL markings, manufacturer details and any configuration limits are present and legible.	☐
Beam condition	Inspect beam section for bending, twisting, dents, corrosion, cracking, excessive deflection, impact damage or signs of overload.	☐
Support points	Check hangers, brackets, anchors, welds, bolts, clamps and parent structure for looseness, corrosion, cracking, movement or distortion.	☐
Trolley / travelling parts	Inspect trolley wheels, axles, side plates, bearings, pins, spacers and adjustment parts for wear, damage, alignment and secure fitting.	☐
Running surface	Check beam flange or track surface for wear, pitting, contamination, obstructions, poor alignment or damage that could affect travel.	☐
End stops and limits	Confirm end stops, buffers, retaining devices, anti-derailment features and travel limits are present, secure and effective.	☐
Load attachment	Check connection points, hooks, shackles, hoists, clamps or lifting accessories for correct type, rating, condition and compatibility.	☐
Operation	Where safe, move the trolley/beam through its travel and check for smooth movement, binding, abnormal noise, excessive play or derailment risk.	☐
Documentation	Confirm drawings, calculations, inspection records, test certificates and defect history are available and current.	☐

6. Removal from service / escalation triggers

- Unknown or illegible SWL/WLL where the beam is used for lifting or suspension.
- Visible cracking, deformation, excessive corrosion, damaged welds or significant impact damage.
- Loose, missing, non-standard or damaged fixings, support brackets, end stops or trolley components.
- Trolley derailment risk, wheel damage, excessive side play, binding or abnormal travel noise.
- Evidence of overload, shock load, unauthorised modification, drilled holes, welded repairs or altered support points.
- Any uncertainty over parent structure suitability, load path, configuration or inspection status.

7. Records and portal use

- Keep this datasheet with the equipment record, manufacturer information and relevant inspection reports.
- Record any site-specific restrictions, such as maximum load position, travel limits, approved hoist type or areas where loads must not be parked.
- Update the portal document if manufacturer information, inspection findings or site limitations change.
- Defects that may affect safe use should be reported, recorded and assessed by a competent person before continued use.

Portal filename

Recommended PDF upload path: documents/travelling_beam_technical_datasheet.pdf

Guidance note: This datasheet is for guidance and record-keeping support only. Always follow manufacturer instructions, site procedures, competent-person advice, risk assessment and any written scheme of examination.